Chapter 20 · Semester 3

WAVE MOTION

Notes by Cikgu Suhaili

10 worked examples 5 simulations Exercises with answers

Waves: introduction

A wave transfers **energy** from one place to another without transferring matter. The particles of the medium only vibrate about their equilibrium positions.

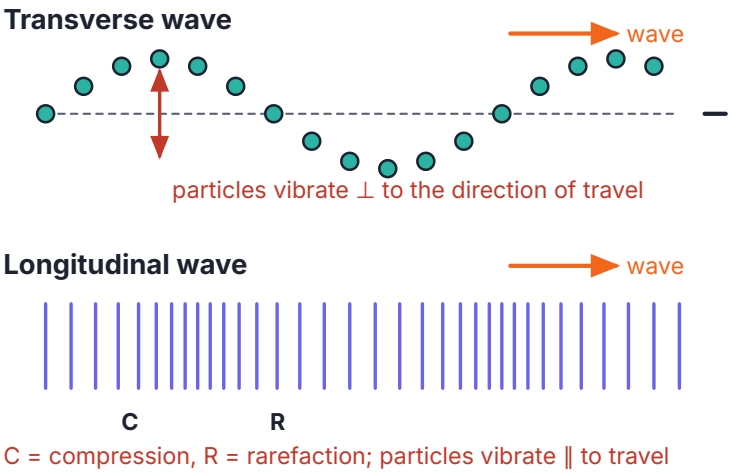


Figure 20.1: transverse and longitudinal waves.

Transverse wave	Longitudinal wave
The particles vibrate **perpendicular** to the direction of propagation.	The particles vibrate **parallel** to the direction of propagation.
crests and troughs	compressions and rarefactions
e.g. waves on a string, water waves, electromagnetic waves	e.g. sound waves, waves along a slinky spring pushed back and forth

20.1 Progressive waves

> **Progressive wave**
> is **a wave in which the wave profile moves through the medium, transferring energy from one point to another**.

Equation of a progressive wave

If the particle at $x = 0$ moves in SHM, $y = A \sin \omega t$, the disturbance reaches a particle at distance x a time x/v later. So the particle at x does the same motion, but later:

$$y = A \sin \omega\left(t - \frac{x}{v}\right) = A \sin \left(\omega t - \frac{\omega}{v} x\right)$$

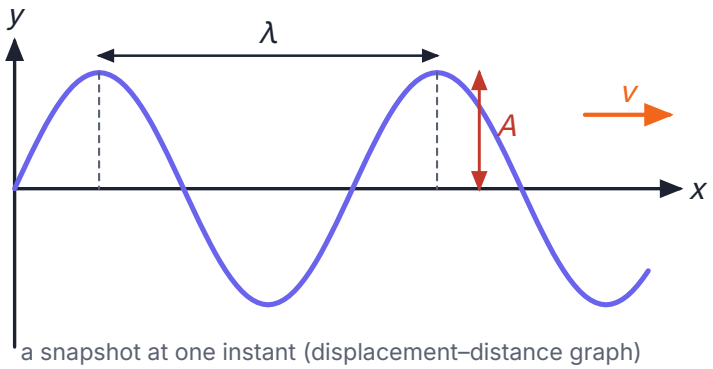
With $k = \frac{\omega}{v} = \frac{2\pi}{\lambda}$ (the wave number):

$$y = A \sin (\omega t - kx) \quad \text{or} \quad y = A \cos (\omega t - kx)$$

$(\omega t - kx)$: wave travelling in the $+x$ direction. $(\omega t + kx)$: wave travelling in the $-x$ direction. A = amplitude, $\omega = 2\pi f$, $k = 2\pi/\lambda$ (rad m^{-1}).

Quantity	Meaning
Wavelength, λ (m)	the distance between two successive points that vibrate in phase (e.g. crest to crest)
Period, T (s)	the time for one complete oscillation of a particle
Frequency, f (Hz)	the number of oscillations per second; $f = 1/T$
Wave number, k (rad m^{-1})	$k = 2\pi/\lambda$
Wave speed, v (m s^{-1})	the distance moved by the wave profile per second

Displacement–distance and displacement–time graphs



a snapshot at one instant (displacement–distance graph)

Figure 20.2: the displacement–distance (y–x) graph is a snapshot of all the particles at one instant. The distance for one cycle is λ.

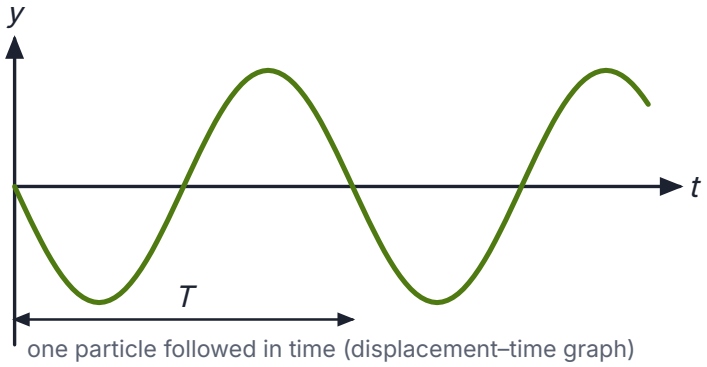


Figure 20.3: the displacement–time (y–t) graph follows one particle. The time for one cycle is T.

Teacher's note

Read the axis first. On a y–x graph, one cycle is the wavelength λ. On a y–t graph, one cycle is the period T. Many students mix these up.

Phase difference

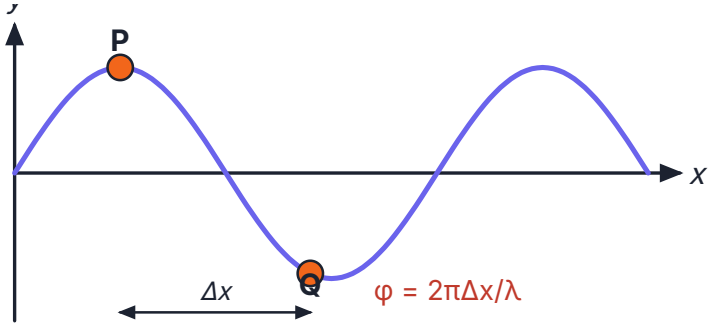


Figure 20.4: two points P and Q a distance Δx apart on the same wave.

At P (x_1): phase = $\omega t - kx_1$; at Q (x_2): phase = $\omega t - kx_2$

phase difference $\varphi = k(x_2 - x_1) = k\Delta x$

$$\varphi = kx = \frac{2\pi x}{\lambda}$$

the phase difference between two points a distance x apart. Points λ apart are in phase ($\varphi = 2\pi$); points λ/2 apart are in antiphase ($\varphi = \pi$).

Wave speed

In one period T, the wave profile moves forward one wavelength λ.

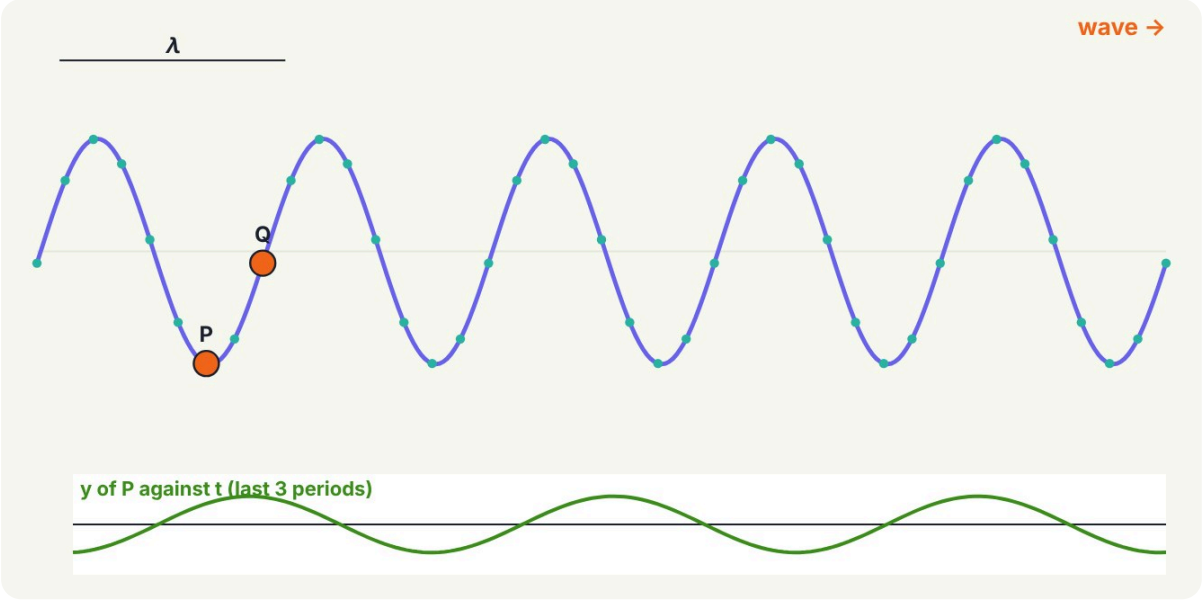
$v = \frac{\lambda}{T} = f\lambda$ (also $v = \frac{\omega}{k}$)

$$v = f\lambda$$

The **wave speed** $v = f\lambda$ is constant in a given medium. The **particle velocity** $\partial y/\partial t = A\omega \cos(\omega t - kx)$ changes all the time, with maximum $A\omega$. Do not mix them up.

Simulation **Progressive wave**

Change the amplitude, wavelength, frequency and direction. Follow the marked particle: it only moves up and down (or back and forth for a longitudinal wave) while the wave profile moves along. The two orange points show the phase difference.



v = fλ	k = 2π/λ	ω = 2πf	T = 1/f
1.00 m/s	**3.14 rad/m**	**3.14 rad/s**	**2.00 s**

phase difference P–Q (2πΔx/λ)	P and Q
1.57 rad = 0.25 cycle	**out of phase**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Example 20.1.1

A progressive wave is represented by $y = 0.05 \sin(8\pi t - 0.4\pi x)$, where x and y are in metres and t in seconds. Find (a) the amplitude, (b) the frequency, (c) the wavelength, (d) the wave speed and its direction, (e) the maximum speed of a particle.

Solution

Compare with $y = A \sin(\omega t - kx)$.

(a) A = **0.05 m** (b) $\omega = 8\pi \Rightarrow f$ = **4 Hz**

(c) $k = 0.4\pi = 2\pi/\lambda \Rightarrow \lambda$ = **5 m**

(d) $v = f\lambda$ = **20 m s^{-1}**, in the **+x** direction (minus sign)

(e) $v_{max} = A\omega = 0.05 \times 8\pi$ = **1.26 m s^{-1}**

Example 20.1.2

A wave of frequency 500 Hz travels at 350 m s^{-1}. (a) What is the distance between two points that are π/3 rad out of phase? (b) What is the phase difference between two displacements at one point 1.0 ms apart?

Solution

$\lambda = v/f = 350/500 = 0.70$ m

(a) $\varphi = 2\pi x/\lambda \Rightarrow x = (\pi/3)(0.70)/(2\pi)$ = **0.117 m**

(b) $\varphi = \omega t = 2\pi(500)(1.0 \times 10^{-3})$ = **π rad**

20.2 Wave intensity

Intensity of a wave

is **the rate of energy transferred (power) per unit area perpendicular to the direction of propagation**.

$$I = \frac{P}{A}$$

unit: W m^{-2}

Intensity and amplitude

The energy of a particle in SHM is $\frac{1}{2}m\omega^2A^2$, so for waves of the same frequency the energy carried is proportional to the square of the amplitude:

$$I \propto A^2$$

doubling the amplitude makes the intensity four times larger

Intensity from a point source

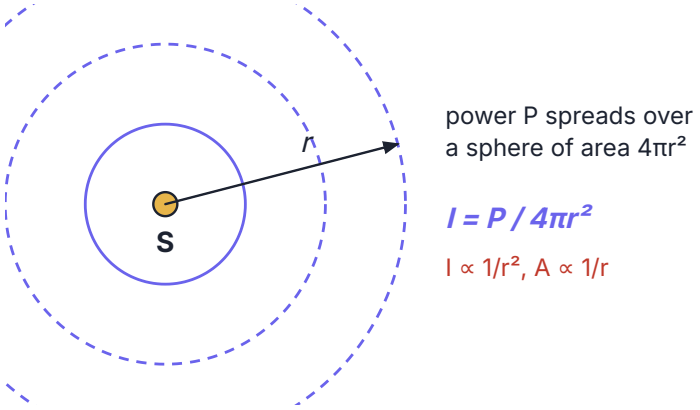


Figure 20.5: a point source radiates power P equally in all directions.

- The energy spreads out over the surface of a sphere of area $4\pi r^2$, so $I = \frac{P}{4\pi r^2}$.
- The intensity therefore decreases with distance: $I \propto \frac{1}{r^2}$ (the inverse square law).
- Since $I \propto A^2$, the amplitude decreases as $A \propto \frac{1}{r}$.

Example 20.2.1

A point source emits sound of power 2.0 W uniformly in all directions. Find (a) the intensity 5.0 m from the source, (b) the ratio of the amplitudes at 5.0 m and at 10.0 m.

Solution

(a) $I = \dfrac{P}{4\pi r^2} = \dfrac{2.0}{4\pi(5.0)^2} = \mathbf{6.4 \times 10^{-3}\ W\ m^{-2}}$

(b) $A \propto 1/r \Rightarrow A_5 : A_{10} = 10 : 5 = \mathbf{2 : 1}$

Example 20.2.2

The amplitude of a wave is reduced to 60% of its original value. What percentage of the original intensity remains?

Solution

$I \propto A^2 \Rightarrow I'/I = (0.60)^2 = 0.36$, i.e. **36%**

20.3 Principle of superposition

Principle of superposition

states that **when two or more waves meet at a point, the resultant displacement at that point is the vector sum of the displacements due to each individual wave**.

$$y = y_1 + y_2$$

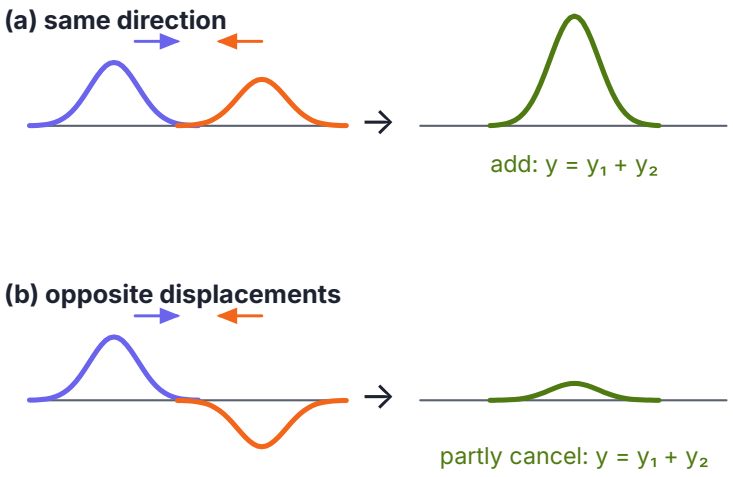
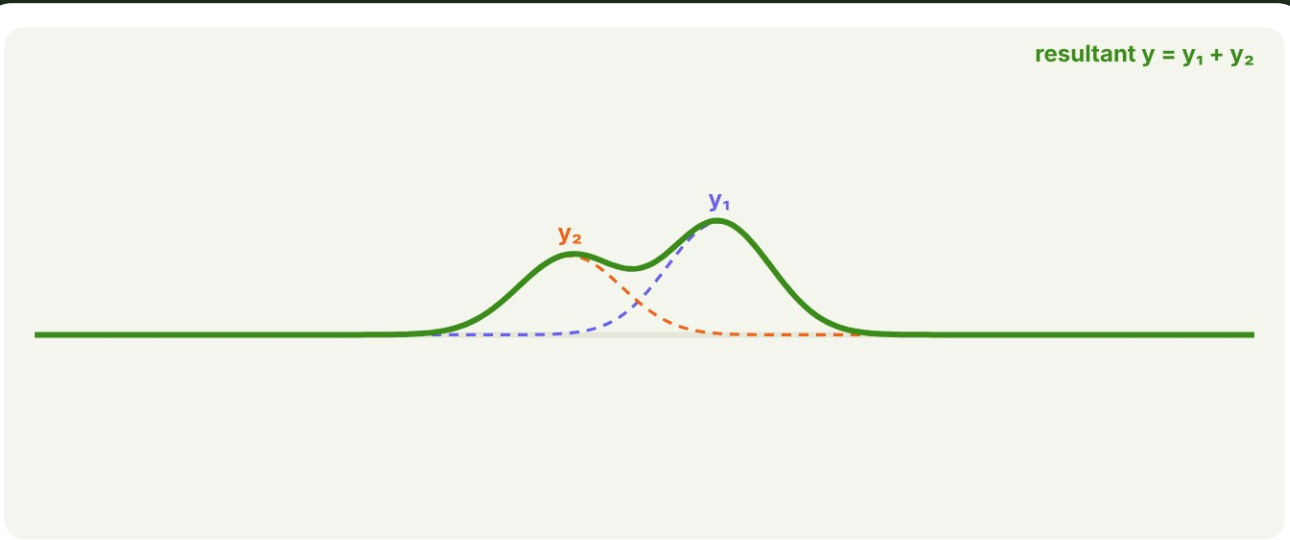


Figure 20.6: two pulses meeting. (a) Displacements in the same direction add. (b) Displacements in opposite directions partly cancel. After meeting, each pulse continues unchanged.

- The waves **interfere** with each other only while they overlap.
- **Constructive interference**: the displacements are in the same direction, giving a larger resultant.
- **Destructive interference**: the displacements are in opposite directions, giving a smaller resultant (zero if they are equal and opposite).

Simulation **Superposition of two pulses**

Send two pulses towards each other, on the same side or on opposite sides. Pause while they overlap: the resultant (green) is the sum of the two displacements at every point. After they pass, each pulse is unchanged.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Example 20.3.1

Two waves $y_1 = 3 \sin(\omega t - kx)$ cm and $y_2 = 4 \sin(\omega t - kx + \pi)$ cm travel along the same string. Find the amplitude of the resultant wave.

Solution

> $\sin(\theta + \pi) = -\sin\theta$, so $y_2 = -4 \sin(\omega t - kx)$
>
> $y = y_1 + y_2 = -1 \sin(\omega t - kx) \Rightarrow$ amplitude **1 cm** (destructive interference)

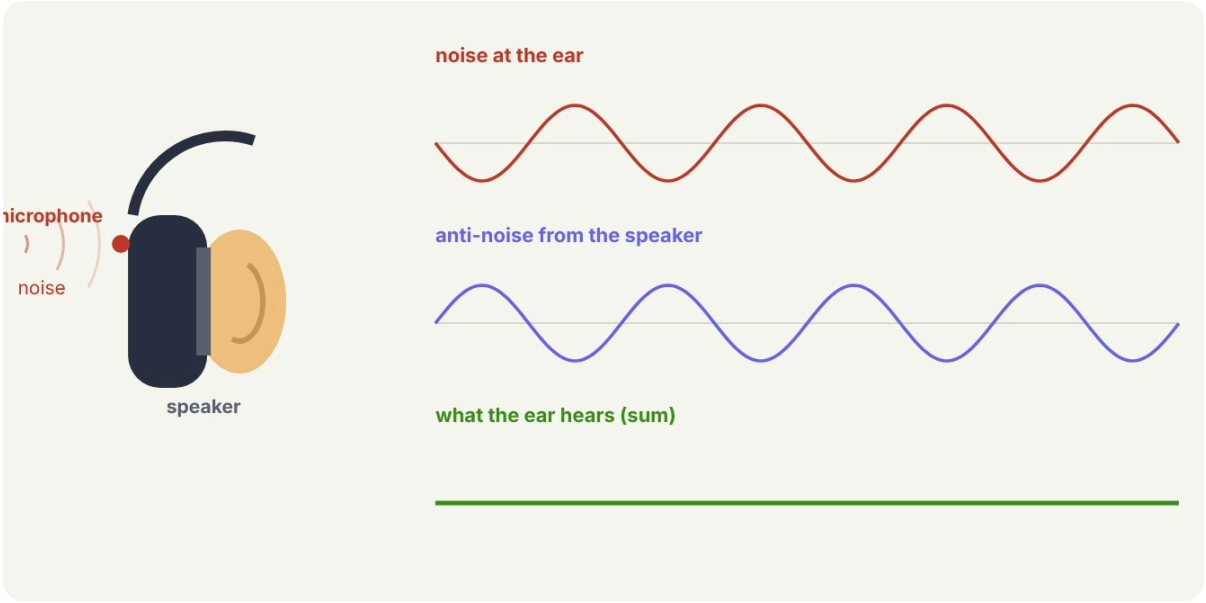
A real use: noise-cancelling headphones

- A small microphone on the outside of each ear cup picks up the noise around you (an aeroplane engine, a bus, an air-conditioner).
- The electronics produce an "anti-noise" wave with the **same amplitude** but **opposite phase** (a phase difference of π), and the speaker plays it into your ear.
- At your ear the two waves superpose: $y = y_{\text{noise}} + y_{\text{anti}} \approx 0$. This is **destructive interference**.
- It works best for steady, low-frequency sounds. For high frequencies even a tiny delay in the electronics is a large fraction of a period, so the anti-noise is no longer in antiphase.

Simulation

Noise-cancelling headphones

Switch the noise cancelling on and off and compare the sound reaching the ear (green). Then add a delay in the electronics or change the noise frequency: the cancellation gets worse once the phase difference moves away from π. Switch on the sound to hear it (use earphones for the low hum).



phase difference between noise and anti-noise
180°

amplitude reaching the ear
0% of the noise

delay as a fraction of one period
0.00 T

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

20.4 Standing waves

> **Standing (stationary) wave**
> is formed by **the superposition of two progressive waves of the same amplitude and frequency, travelling at the same speed in opposite directions**.

Formation by reflection

A wave travelling along a string is reflected at the end. The incident and reflected waves travel in opposite directions and superpose to form a standing wave.



Figure 20.7: a pulse is inverted when it is reflected at a fixed end, and is not inverted at a free end.

- At some points the two waves are always equal and opposite, so they always cancel: these points do not move. They are **nodes** (N).
- Midway between the nodes, the two waves always add, giving the largest oscillation. These points are **antinodes** (A).

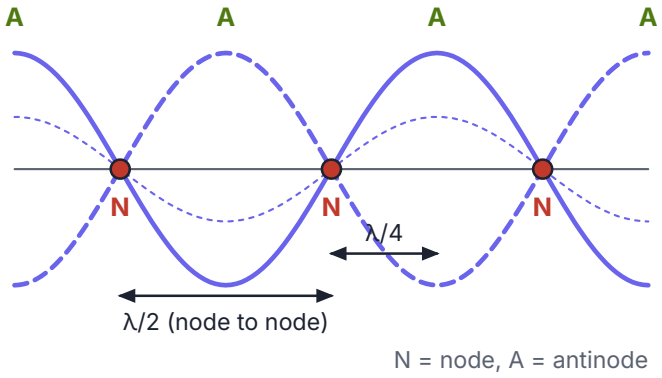


Figure 20.8: a standing wave. The profile does not move along; each point oscillates with its own amplitude.

The standing wave equation

$y_1 = a \sin(\omega t - \frac{2\pi x}{\lambda})$ (to the right), $y_2 = a \sin(\omega t + \frac{2\pi x}{\lambda})$ (to the left)

$y = y_1 + y_2$, using $\sin P + \sin Q = 2 \sin \frac{P+Q}{2} \cos \frac{P-Q}{2}$

$y = 2a \sin \omega t \cos(-\frac{2\pi x}{\lambda}) = 2a \cos \frac{2\pi x}{\lambda} \sin \omega t$

$$y = 2a \cos \frac{2\pi x}{\lambda} \sin \omega t = (2a \cos kx) \sin \omega t$$

every point oscillates in SHM ($\sin \omega t$) with an amplitude $A = 2a \cos kx$ that depends on its position x

Other books write $y = 2A \sin kx \cos \omega t$. It is the same standing wave with the origin at a node instead of an antinode.

Nodes: amplitude = 0

$\cos \frac{2\pi x}{\lambda} = 0 \Rightarrow \frac{2\pi x}{\lambda} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \ldots = \frac{n\lambda}{4}, n = 1, 3, 5, \ldots$

distance between successive nodes $= \frac{3\lambda}{4} - \frac{\lambda}{4} = \boldsymbol{\frac{\lambda}{2}}$

Antinodes: amplitude = $2a$ (maximum)

$\cos \frac{2\pi x}{\lambda} = \pm 1 \Rightarrow \frac{2\pi x}{\lambda} = 0, \pi, 2\pi, \ldots$

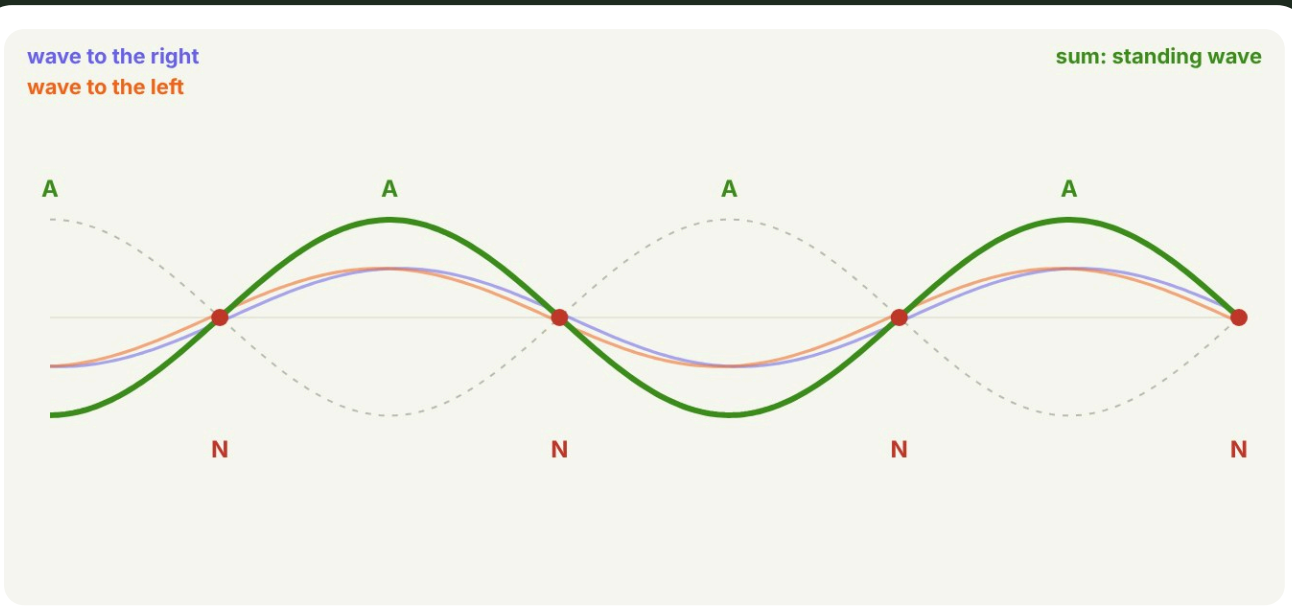
$x = 0, \frac{\lambda}{2}, \lambda, \ldots = \frac{n\lambda}{4}, n = 0, 2, 4, \ldots$

distance between successive antinodes $= \boldsymbol{\frac{\lambda}{2}}$;

node to next antinode $= \boldsymbol{\frac{\lambda}{4}}$

Simulation Forming a standing wave

Two identical waves travel in opposite directions (blue and orange). Their sum (green) is a standing wave: the nodes never move and the antinodes have the largest amplitude. Switch to the string to see the harmonics of a string fixed at both ends.



node to node	node to antinode	amplitude at an antinode	amplitude at a node
λ/2	**λ/4**	**2a**	**0**

This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Standing waves on a stretched string

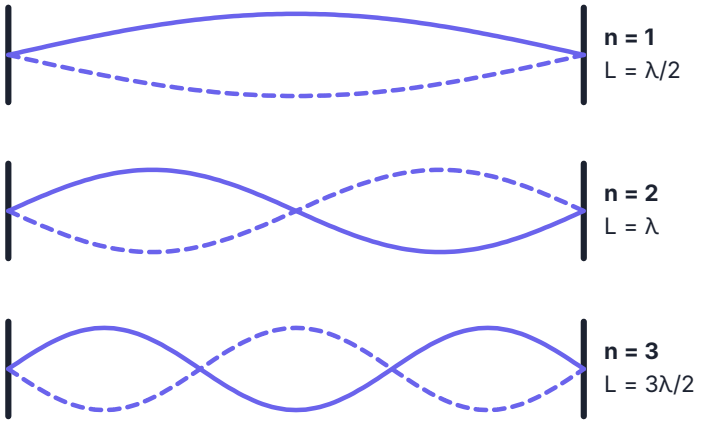


Figure 20.9: a string fixed at both ends has a node at each end. Only waves that fit a whole number of half-wavelengths can form.

- The lowest frequency (the **fundamental frequency** or **first harmonic**): L = λ/2, so λ = 2L and f_1 = v/2L.
- Next: L = λ, the second harmonic (first overtone), f_2 = 2f_1.
- In general L = nλ/2 and f_n = nv/2L. More about strings and pipes in Chapter 21 Sound waves.

Standing waves and progressive waves

Standing wave	Progressive wave
The wave profile does not move; energy is not transferred from one particle to the next.	The wave profile moves in the direction of propagation and energy is transferred.
There are points (nodes) that are not displaced at all.	No nodes.
Midway between two successive nodes is an antinode, where the amplitude is maximum.	No antinodes.
Points between a node and an antinode have different amplitudes of oscillation.	All points have the same amplitude of oscillation.
Points between two successive nodes oscillate in phase, with the same frequency.	Neighbouring points are not in phase, but have the same frequency.

Example 20.4.1

A standing wave is represented by $y = 2\cos 4x \sin 6t$ (SI units). Calculate (a) the frequency, (b) the wavelength, (c) the distance between successive nodes, (d) the amplitude at $x = 0.50$ m.

Solution

Compare with $y = 2a\cos\dfrac{2\pi x}{\lambda}\sin\omega t$.

(a) $\omega = 6 = 2\pi f \Rightarrow f =$ **0.955 Hz**

(b) $\dfrac{2\pi}{\lambda} = 4 \Rightarrow \lambda =$ **1.57 m**

(c) $\dfrac{\lambda}{2} = \dfrac{1.57}{2} =$ **0.785 m**

(d) $A = 2\cos 4(0.50) = 2\cos 2.0 = -0.832$; amplitude = **0.832 m** (calculator in radians; the amplitude is the magnitude)

Example 20.4.2

A standing-wave pattern is observed in a thin wire of length 3.00 m. The equation of the wave is y = (0.002 m) sin (πx) cos (100πt), with x in metres and t in seconds. (a) How many loops does the pattern have? (b) What is the frequency? (c) What is the distance between adjacent nodes?



Solution

Compare with $y = 2A \sin kx \cos \omega t$.

$k = 2\pi/\lambda = \pi \Rightarrow \lambda = 2.00$ m; each loop is $\lambda/2 = 1.00$ m long

(a) number of loops = 3.00/1.00 = **3**

(b) $\omega = 2\pi f = 100\pi \Rightarrow f =$ **50.0 Hz**

(c) $\lambda/2 =$ **1.00 m**

Example 20.4.3

The speed of waves in a guitar string is 425 m s^{-1}. Find the fundamental frequency (first harmonic) of the string if its length is 76.5 cm.

Solution

$\lambda = 2L = 2(0.765) = 1.53$ m

$f = v/\lambda = 425/1.53 =$ **278 Hz**

Example 20.4.4

Find the length of guitar string needed to produce a fundamental frequency of 256 Hz, if the speed of waves in the string is 405 m s^{-1}.

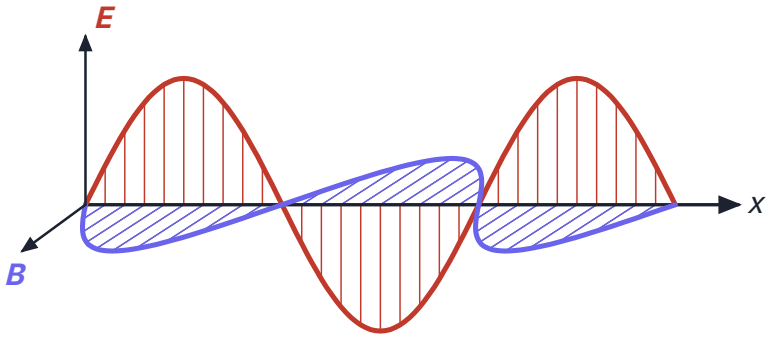
Solution

$\lambda = v/f = 405/256 = 1.58$ m

$L = \lambda/2 =$ **0.791 m**

20.5 Electromagnetic waves

An electromagnetic (EM) wave is made up of an **electric field** and a **magnetic field** that vibrate perpendicular to each other and to the direction of propagation, in phase.



E ⊥ B ⊥ direction of travel; E and B in phase

Figure 20.10: an electromagnetic wave travelling in the +x direction.

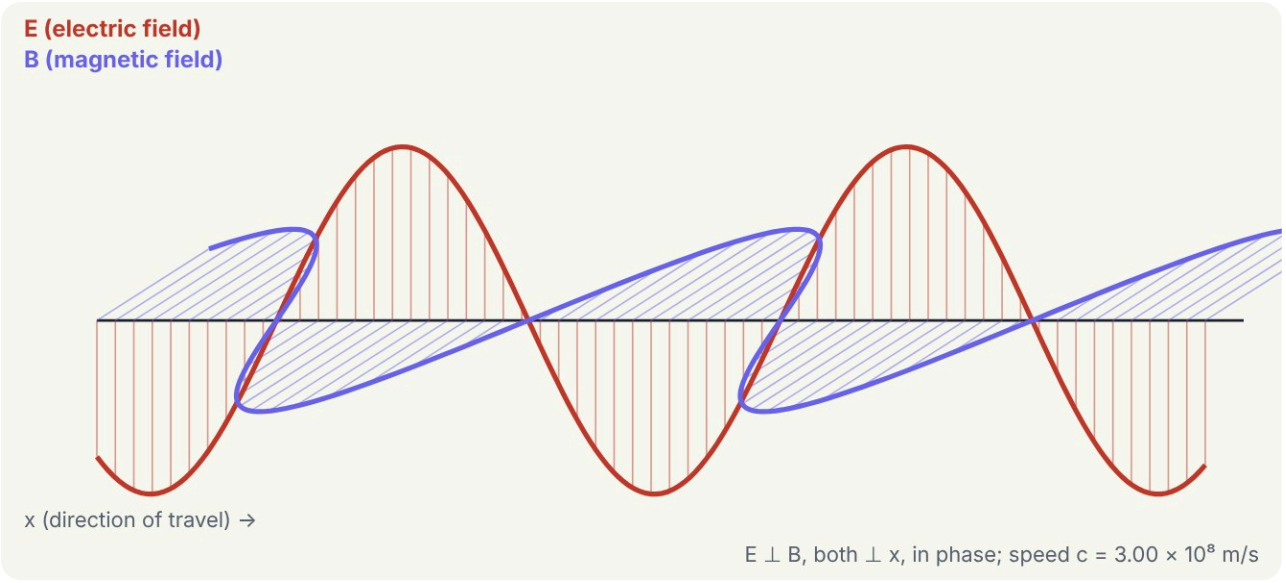
$$E = E_0 \sin(\omega t - kx) \quad B = B_0 \sin(\omega t - kx)$$

E_0, B_0 = amplitudes of the electric and magnetic fields; both have the same ω and k, so they are in phase

Simulation

Electromagnetic wave

The electric field (red) and the magnetic field (violet) vibrate at right angles to each other and to the direction of travel, and reach their maxima together.



This is a still picture. Open PhysiQ › Notes › Semester 3 › Chapter 20 to use the simulation.

Characteristics of electromagnetic waves

- They are transverse waves.
- They do not need a medium: they can travel through a vacuum.
- They all travel at the same speed in a vacuum, $c = 3.00 \times 10^8\ \text{m s}^{-1}$.
- They carry no charge, so they are not deflected by electric or magnetic fields.
- They show reflection, refraction, interference, diffraction and polarisation.
- They obey $c = f\lambda$.

Electromagnetic waves and mechanical waves

Electromagnetic waves	Mechanical waves
vibrations of electric and magnetic fields	vibrations of the particles of a medium
do not need a medium; can travel in a vacuum	need a medium; cannot travel in a vacuum
always transverse	transverse or longitudinal
speed in a vacuum = 3.00×10^8 m s^{-1}	much slower (e.g. sound in air ≈ 340 m s^{-1})
e.g. light, radio waves, X-rays	e.g. sound, water waves, waves on a string

The speed of light

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} (permeability of free space); $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} (permittivity of free space)

Significance: the speed of EM waves can be calculated from two constants of electricity and magnetism: $1/\sqrt{(4\pi \times 10^{-7} \times 8.85 \times 10^{-12})} = 3.00 \times 10^8$ m s^{-1}, which is the measured speed of light. This showed that **light is an electromagnetic wave**.

The electromagnetic spectrum

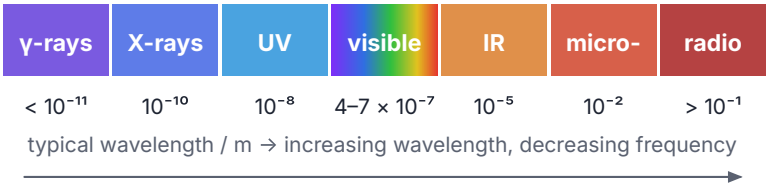


Figure 20.11: the electromagnetic spectrum.

Radiation	Typical wavelength / m	Typical frequency / Hz
γ-rays	$< 10^{-11}$	$> 10^{19}$
X-rays	$10^{-11} – 10^{-8}$	$10^{16} – 10^{19}$
ultraviolet	$10^{-8} – 4 \times 10^{-7}$	$10^{15} – 10^{16}$
visible light	$4 \times 10^{-7} – 7 \times 10^{-7}$	$\approx 10^{14} – 10^{15}$
infrared	$7 \times 10^{-7} – 10^{-3}$	$10^{11} – 10^{14}$
microwaves	$10^{-3} – 10^{-1}$	$10^{9} – 10^{11}$
radio waves	$> 10^{-1}$	$< 10^{9}$

Teacher's note

For the exam, remember the **order** of the spectrum and the order of magnitude of each region. The boundaries are not sharp, so books give slightly different values.

Example 20.5.1

The electric field of an EM wave in a vacuum is $E = 150 \sin (6.0 \times 10^{15} t - kx)$ V m^{-1}. Find (a) the frequency, (b) the wavelength, (c) the region of the spectrum.

Solution

(a) $f = \omega/2\pi = 6.0 \times 10^{15}/2\pi =$ **9.5×10^{14} Hz**

(b) $\lambda = c/f = 3.00 \times 10^{8}/9.5 \times 10^{14} =$ **3.1×10^{-7} m**

(c) **ultraviolet**

Summary and exercises

Subtopic	Key results
20.1	$y = A \sin(\omega t \pm kx)$; $k = 2\pi/\lambda$; $\varphi = 2\pi x/\lambda$; $v = f\lambda$
20.2	$I = P/A$; $I \propto A^2$; point source $I = P/4\pi r^2$
20.3	$y = y_1 + y_2$
20.4	$y = 2a \cos kx \sin \omega t$; node to node = $\lambda/2$; node to antinode = $\lambda/4$
20.5	$E = E_0 \sin(\omega t - kx)$, $B = B_0 \sin(\omega t - kx)$; $c = 1/\sqrt{(\mu_0 \varepsilon_0)}$

Exercises

1. A wave on a string is described by $y = 0.02 \sin(30t + 4x)$ (SI units). Find the wavelength, the frequency, and the speed and direction of the wave.

   Answer

   $\lambda = 1.57$ m, $f = 4.77$ Hz, $v = 7.5\ \text{m s}^{-1}$ in the $-x$ direction

2. Two points on a progressive wave of wavelength 1.2 m are 0.30 m apart. What is the phase difference between them?

   Answer

   $\pi/2$ rad

3. The intensity of a wave 2.0 m from a point source is 0.80 W m^{-2}. What is the intensity 8.0 m from the source?

   Answer

   0.050 W m^{-2}

4. A standing wave on a string has nodes 0.25 m apart. The wave speed is 100 m s^{-1}. Find the frequency.

   Answer

   $\lambda = 0.50$ m, $f = 200$ Hz

5. State two differences between an electromagnetic wave and a sound wave.

Answer

e.g. EM waves can travel through a vacuum but sound needs a medium; EM waves are transverse but sound is longitudinal; EM waves travel at 3.00×10^8 m s^{-1} in a vacuum, much faster than sound.